QUANT AUDIT · STRATEGY REVIEW · JUN 2026

ZeroDTE 0DTE Engine

Current architecture, the trading plan, and the flaws that matter — with a prioritized fix plan.

Prepared as a senior 0DTE index-options quant. Findings are code-verified and adversarially reviewed. Where the edge is unproven, this report says so plainly.

EXECUTIVE SUMMARY

The verdict, up front

› **The engine is well-built; the EVIDENCE behind it is not.**

Clean architecture, real gates, honest backtest. But three things quietly invalidate the 'we're validating an edge' story.

› **1 · You are trading the wrong side of your own edge.**

The validated +$5,479 is ~96% PUT-selling. Live has been ~100% CALL-selling — the side with 6 backtested trades and a structural headwind.

› **2 · The 'live validation' is the model grading its own homework.**

Every live P&L number is a Black-Scholes mid-price simulation. The broker's real fill price is never read. You are validating the model against the model.

› **3 · The edge is statistically marginal and cost-fragile.**

5-year backtest t-stat = 1.85 (below the 1.96 bar). 84% of profit came from 2022–23. It goes negative at ~$45/spread of cost in recent years.

› **Bottom line**

Do not scale. Do not read the green number as proof. Fix the measurement, split the book by side, and re-validate honestly.

ARCHITECTURE

Current architecture — how a trade is born

- ### Source of truth

  Persistent FastAPI backend (:8765). Feed: Alpaca SPY IEX ×10 → IBKR → yfinance. Telegram, the Pine indicator, and the PWA are parallel projections.

- ### Signal engine

  predictor.run_backtest() rolls a 50-bar 5-min buffer through a Pine-faithful detector. A 09:30–09:45 ET observation window classifies regime (VOLATILE if obs_range/ATR > 1.5 → that day trades nothing).

- ### Trigger (09:45–14:00 ET)

  A Stochastic %K-vs-%D reversal cross OUT of an 80/20 extreme. Overbought cross-down → sell-call; oversold cross-up → sell-put.

- ### 8 hard gates

  confluence ≥ 3/4 · VWAP alignment · trend filter · per-side cooldown · max 3 trades/day · max 3 concurrent · VIX bucket · macro blackout · mid-session vol re-gate · 2%/day loss limit.

- ### Pricing & exits

  Strikes at 30Δ via Black-Scholes bisection (flat vol = realized 5-min σ × 1.20). $10 SPX wing. TP at 90% of credit. No stop-ladder. Time-stop 15:30 ET.

- ### Execution

  SPX → SPY at 1/10 scale ($1 grid). MARKET multi-leg orders to Alpaca paper.

CONFIGURATION

The live configuration (verified in .env)

ACCOUNT	RISK / TRADE	MAX TRADES/DAY	NO NEW ENTRY AFTER
$10,000	4% = $400	3	14:00 ET
SHORT DELTA	WING	TAKE-PROFIT	P&L MODEL
~30Δ	$10 SPX	90% credit	BS-mid

Sizing reality: $400 budget vs ~$650 max-loss/contract → recommend_contracts floors to exactly 1 contract every trade. So the GEX size-trim and the dollar-based concurrency controls are structurally inert — they can never fire at this account size.

THE CORE ISSUE

How the system picks CALL vs PUT — and why it matters

THE RULE (predictor.py:388–413)

- Sell-CALL fires on an overbought cross-down, UNLESS trend == 'up'.
- Sell-PUT fires on an oversold cross-up, UNLESS trend == 'down'.

- Trend = EMA10–EMA30 spread vs ±0.05%. Inside that band = 'flat' → BOTH sides fire.
- In a quiet up-drift the stoch hits OVERBOUGHT far more than oversold → the trigger manufactures CALL signals.

- A gentle grind reads 'flat', so the trend filter never blocks those counter-trend calls.

THE CONSEQUENCE

- Validated edge = sell-PUTS-on-dips (with the market's upward drift + fear premium). Robust.
- Live book = sell-CALLS-on-rallies. Works only in genuine down/chop; loses in a stealth grind-up.

- Backtest: 147 puts (+$5,357) vs 6 calls (+$121). The call book is essentially un-tested.
- → Live is trading the unvalidated, structurally-weak side, in the regime that punishes it.

STATISTICAL REALITY

What the headline number actually proves

5YR P&L	PER-TRADE	PER-TRADE Σ	T-STAT
+$5,479	+$35.8	$240	1.85

95% BAR	CLEAN TRADES TO CONFIRM	LIVE TRADES SO FAR	FROM 2022–23
t > 1.96	~173	~4	84%

Even five years of backtest does NOT clear the 95% significance bar. The mean edge is small relative to the noise, and the bulk of it is a 2022–23 bear-market put-selling artifact. A handful of paper trades cannot possibly confirm this.

THE FLAWS

Where it actually costs money

STRATEGY

Strategy — the put / call inversion

Live trades the CALL book; the validated edge is 96% PUTS

CRITICAL

Sell-calls-on-rallies is counter-trend in an up-drift and has only 6 backtested trades (+$121) behind it. The +$5,479 is the put book. The trend filter's 0.05% 'flat' band lets counter-trend calls through in a stealth grind.

FIX Split ALL reporting into put-book vs call-book (done). Stage a flag to suppress / down-size the call book pending a per-side backtest. Re-frame the strategy honestly as put-driven.

Strikes priced on a FLAT volatility surface — no skew

HIGH

tv = realized 5-min σ × 1.20 is symmetric. Real 0DTE puts carry a rich skew premium; calls are cheaper. A flat surface over-credits calls and under-credits puts — flattering exactly the side you shouldn't be selling.

FIX Add a two-parameter skew (put-IV uplift, call-IV discount) calibrated to sampled real chains; re-run the backtest before trusting any call-side attribution.

COST / VALIDATION

Measurement — the validation is circular

P&L is a Black-Scholes mid-price simulation grading itself

CRITICAL

Every live pt.pnl = estimated_credit × exit_pct – $25, with both credit and exit = bs.spread_value() (theoretical mid). A full grep confirms filled_avg_price / avg_price are NEVER read. The 'shadow validation' and the backtest are two outputs of the SAME kernel — it cannot observe slippage at all.

FIX Read Alpaca filled_avg_price per leg → store broker_realized_pnl SEPARATELY from model_pnl. Judge the 4–6-week validation on broker-realized P&L only.

MARKET multi-leg orders guarantee worst-case slippage

HIGH

alpaca_trader.py submits type='market' mleg orders. On 0DTE wings the bid/ask is wide; market orders cross the full spread on entry AND exit. The model assumes mid on both — so realized cost is structurally worse than the $25 the backtest charges.

FIX Move to marketable-limit at mid (1 tick give), short reprice loop, market only as a time-stop fallback. Add a liquidity gate once a live NBBO feed exists.

BACKEND

Backend logic — the live path doesn't match the backtest

Exits evaluate on the DEVELOPING 5-minute bar → phantom wins

HIGH

The feed re-dispatches the still-forming bar every 60s and _check_open_wave_trades runs on every dispatch with no bar-closed guard. honest_backtest only acts on CLOSED bars (worst-first). Live can book an intra-bar TP the backtest would never see — corrupting the small sample optimistically.

FIX Gate exit evaluation to closed bars only (act on bar N when a newer timestamp arrives). Keep intra-bar dispatch for quotes/UI only.

Volatility is frozen at entry — blind to intraday IV expansion

HIGH

bs_realized_std is captured once and only time-to-expiry shrinks. On an afternoon news-pop the frozen-low vol under-prices the loss side: the model says −60% while you're really near +110% and breaching. It mis-prices exactly the fat-tail days.

FIX Add a conservative vol-floor ratchet (lift tv when rolling realized vol exceeds entry; never reduce a loss estimate). Re-validate.

Restart / outage holes corrupt the small-sample ledger

HIGH

Signal keys are in-memory only; a mid-session restart re-dispatches the day's bars and can re-book phantom duplicate trades (burning the 3/day cap, injecting fake P&L). Prior-day 0DTE positions aren't force-settled on startup before the day-gate discards them.

FIX Persist signal keys + a hard bar-age guard so backfilled signals never open trades; force-settle open prior-date 0DTE on startup; append-only ledger (done).

MACRO / TIMING

Macro & timing — the backtest and the live system are different animals

Backtest has ZERO event awareness; live blocks event windows

HIGH

honest_backtest takes every signal; live enforces a hard macro blackout. 22% of the validated exits land on FOMC/CPI/NFP/OPEX days and were MORE profitable in-sample. So live blocks the days that disproportionately built the edge — the headline number doesn't describe how it actually trades.

FIX Re-run the backtest with a macro-calendar overlay: 'event-days-included' vs 'event-days-blocked' P&L. The single most valuable missing number.

Dealer-gamma (GEX) is collected but never acted on — and is inert at 1-contract sizing

HIGH

Negative GEX = vol-amplified = the short strike is likelier to breach before TP. The system stamps the regime for logging and trades full size into it. The only implemented response is a size-trim that cannot fire when every trade is 1 contract.

FIX Measure breach rate on negative vs positive GEX days in the backtest first; if adverse, use GEX as a STAND-ASIDE / IC-only gate (not a size-trim).

Event-day EXIT costs unmodeled; macro gate can fail silently

MED

Both engines charge a flat $25 on FOMC as on a quiet Tuesday; open positions exit into 3–5× wider spreads at the modeled mid. A 3× closing haircut drops the validated total to $4,629 (t→1.56); 5× to $3,779 (t→1.28). The blackout also depends on impact-string labels that can drift (e.g. a missing PCE).

FIX Add an event-day cost multiplier to the backtest; emit a morning log of exactly which events were classified high-impact and their blackout windows.

COST SENSITIVITY

The edge curve — where it dies

$25 / SPREAD +$5,479	$35 / SPREAD +$3,949	$45 / SPREAD ≈ $0 *	$60 / SPREAD +$124
2022–23 SHARE 84%	RECENT-REGIME P&L ~$678	LIVE BREAKEVEN (SPY) ~$6/spr	ORDER TYPE MARKET

* At $45/spread the 2024, 2025 and 2026 years are individually negative. Realistic 0DTE costs are $15–40 + market-order slippage on top. The edge sits INSIDE the cost-assumption error bar — and the regime you actually trade in now contributed almost none of it.

THE FIX PLAN

Recommendations — in priority order

› 1 · Confront the put/call inversion

Report per-side. Stop reading the +$5,479 as evidence for the call book. Stage call-suppression / down-sizing pending a per-side backtest.

› 2 · Wire REAL fills into the ledger before any deploy decision

Read filled_avg_price; store broker_realized_pnl beside model_pnl; judge validation on broker-realized only. Move off market orders.

› 3 · Fix the developing-bar exit bug

Gate exits to closed bars so the live path mirrors the backtest it claims to validate.

› 4 · Run the event-overlay backtest

Event-included vs event-blocked P&L; tag trades by GEX regime and OPEX. Cheapest high-value number.

› 5 · Stress cost & demand significance

Publish the $25/$40/$50/$61 curve in every debrief. Do not scale contracts until it survives $50+ AND ~100 clean broker-realized trades.

› 6 · Close the restart / expiry data holes

Bar-age guard; force-settle prior-day 0DTE on startup; append-only ledger. One lost ITM loser distorts a 4-trade sample.

› 7 · Price the tails honestly

Conservative vol-floor ratchet so intraday vol expansion lifts the loss side.

› 8 · Act on GEX or stop claiming it as a safety control

Validate negative-vs-positive breach rates, then use GEX as a stand-aside gate — or drop the claim.

CHANGE CONTROL

What I changed tonight vs what I staged for your sign-off

DEPLOYED NOW (safe · no edge change)

- Put-book vs call-book split in the debrief + EOD.
- Cost-sensitivity edge curve ($25→$61) in the backtest + debrief anchors.
- Append-only persisted trade ledger (never date-gated).
- Broker-fill instrumentation scaffold (model_pnl vs broker_realized_pnl).
- Config flags added, defaulting to CURRENT behaviour.

STAGED (flag, OFF by default — your call)

- DIRECTIONAL_SUPPRESS_CALLS — kill / down-size the call book.
- EXIT_ON_CLOSED_BAR_ONLY — fix the developing-bar exits.
- Skew-aware pricing; vol-floor ratchet (re-validate first).
- Marketable-limit execution; event-day timing + cost model.
- GEX stand-aside gate; tighter trend filter.

CLOSING

The bottom line

- **You have not yet proven anything — and that's fine, if you measure honestly.**
  The architecture is sound. The problem is the instrument panel is wired to the model, not the market.

- **Three numbers to live by from here:**
  broker-realized P&L (not model P&L) · per-side edge (puts vs calls) · the cost-curve breakeven. Everything else is noise.

- **Do this before risking a cent of real money:**
  ≥ ~100 clean, broker-realized, closed-bar trades · positive at $50+/spread cost · put/call mix consistent with the backtest · drawdown inside –$1,581.

- **The discipline that makes money here is subtraction:**
  fewer trades, the right side, real fills, and the patience to let the sample speak. The edge — if it's real — is a put-selling edge. Trade that one.